Md. Taifur Hossain.

CS UNDERGRADUATE · 3D VISION & ML SYSTEMS

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Final-year Computer Science undergraduate focused on memory-efficient 3D Gaussian Splatting. Built and released four open-source tools across 3D reconstruction, model compression, and ML tooling, plus a full multi-scene research pipeline. Comfortable taking a problem from experimentation through to a tested, documented release.

EDUCATION

International University of Business Agriculture & Technology (IUBAT) — Dhaka, BD

B.Sc. in Computer Science & Engineering (BCSE) — Expected 2027

CGPA 3.4 / 4.00

PUBLICATIONS

Beyond Naive INT8: Adaptive Post-Training Compression for 3D Gaussian Splatting · ACCEPTED

M. T. Hossain, S. S. Sourav, M. A. Hossain. Proc. ICCA 2026 (ACM), Dhaka.

Designed a training-free post-processing pipeline (adaptive pruning → outlier cleaning → spherical-harmonic reduction → mixed-precision quantization) achieving a **74% mean size reduction** across three scenes at a 1.3 to 2.7 dB PSNR cost.

Diagnosed a per-column INT8 quantization collapse (~15 dB uniform) and recovered to within **0.1 to 0.7 dB of FP16** using Morton Z-order sub-group ranges, a positive and reproducible quantization finding.

Isolated a zero-update failure mode in post-pruning fine-tuning (optimizer-state dimension mismatch), documented as a reproducible negative result.

Ran the full multi-scene pipeline on a single 16 GB Kaggle T4 GPU using the Nerfstudio / gsplat ecosystem.

PROJECTS

splat-slim
Python · Typer · NumPy · CI · repo

Training-free CLI / library that compresses trained 3DGS `.ply` files ~74% with <3 dB PSNR loss; ships self-decodable Morton-ordered sub-group INT8 with per-group ranges in a sidecar `.meta.npz`.

splat-lab
Vite · TypeScript · Three.js · live demo

In-browser, real-time demo of the compression pipeline with opacity / scale / SH / quantization sliders and an honest on-screen size readout; a TypeScript port of the splat-slim math.

nerf-prep
Python · Docker · GitHub Actions · repo

One-command Dockerized COLMAP → Nerfstudio dataset prep: automatic matcher selection by capture type, GPU/CPU auto-detect, headless operation, input validation, and a registration-rate summary.

paper-chat
Python · Streamlit · sentence-transformers · repo

RAG over a PDF / arXiv library with per-(paper, page) citations, cross-paper questions, and grounded-only answers (declines when sources don't cover the question).

lumina-lms
Next.js · React · TypeScript · Prisma · repo

Full-stack AI learning management system: a retrieval-augmented tutor that answers from course materials and cites its sources, role-based access for students, instructors and admins, adaptive learning paths, gamification and analytics dashboards.

medical-expert-system
Python · Flask · scikit-learn · repo

Hybrid diagnosis expert system combining rule-based confidence factors, A* and Best-First search, Bayesian inference and a Decision Tree plus MLP classifier in a weighted ensemble, served through a Flask interface.

TECHNICAL SKILLS

LANGUAGES	Python, C, C++, C#, TypeScript / JavaScript, Bash, LaTeX
ML & 3D VISION	3D Gaussian Splatting, NeRF, Nerfstudio, gsplat, PyTorch, scikit-learn, COLMAP, model quantization, Bayesian inference, RAG & embeddings
TOOLS & INFRA	Next.js, React, Prisma, Flask, Streamlit, Three.js, Vite, Docker, Git, GitHub Actions (CI/CD), Linux